3.6.1 Battery management

Many battery-powered sensors promise multi-year operation, but that assumes ideal conditions. In reality, batteries drain faster in temperature extremes, with frequent transmissions, or poor network signal quality. Without systematic monitoring, the first indication of a dead battery is missing data only noticed days or weeks later. In my deployment, I've implemented battery voltage reporting in every sensor message, allowing me to predict when batteries will run out and replace them before they do (almost a meta digital twin of the sensor network!). Industrial deployments often standardize on specific battery types and implement replacement schedules based on worst-case scenarios rather than manufacturer estimates.

3.6.2 Firmware updates

Security vulnerabilities, bug fixes, and feature additions require firmware updates throughout a sensor's lifetime. The limited bandwidth of LPWAN networks makes transmitting firmware images challenging, battery-powered devices must manage updates without draining power, failed updates can brick remote sensors requiring expensive site visits, and heterogeneous networks mean different update procedures for each device type. My LoRaWAN sensors require updates during maintenance windows, transmitting firmware in small chunks over days, and maintaining rollback capabilities for failed updates.

3.6.3 Physical asset tracking

Sensors transmit streams of data as soon as they are powered on. Without any knowledge of where that sensor is, or what it is measuring, that data stream is meaningless or misleading. I maintain both logical locations (what the sensor measures), and the physical location where each sensor is, combined with photographs to show exact mounting positions.

3.6.4 Calibration drift

The edge processing camera I have installed on my water meter requires careful and precise calibration to ensure the ML model extracts the correct digits from the image. Any slight movement on the camera mounting requires